KONGU ENGINEERING COLLEGE

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**DEPARTMENT OF INFORMATION TECHNOLOGY**

**24ITT32 - PROGRAMMING IN JAVA**

# INDIVIDUAL PROJECT ON ACADEMIC MANAGEMENT SYSTEM

**PROJECT REPORT**

Submitted by

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# JAVA ASSIGNMENT

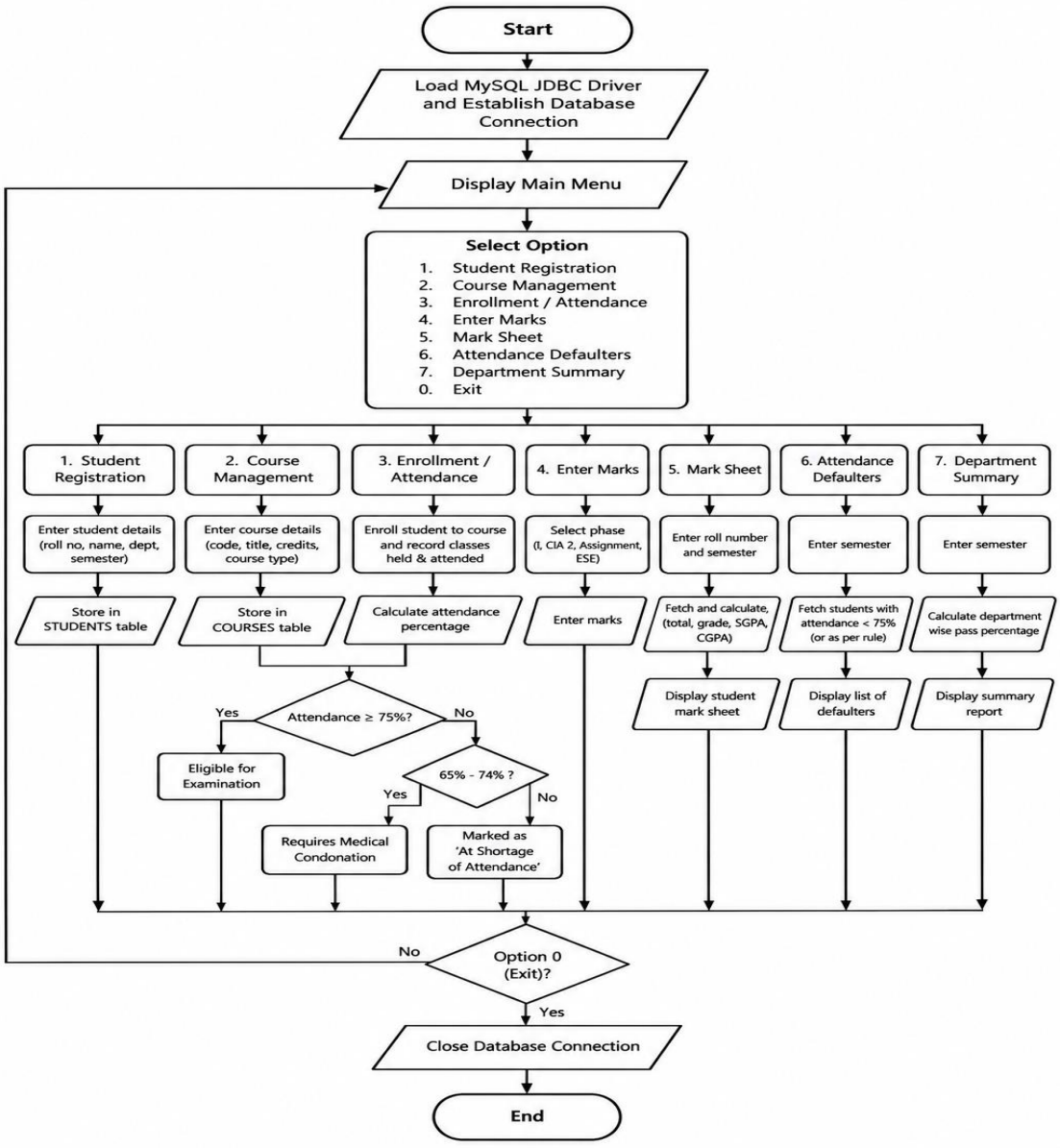
## AIM:

To develop a Java JDBC and MySQL-based academic management application for maintaining student academic information and supporting continuous evaluation workflows.

## OBJECTIVE:

- Store and retrieve student records including roll number, student name, department, and semester.
- Maintain course information such as course code, course title, credits, and course type.
- Enroll students in courses and manage attendance information, including medical-condonation updates.
- Support phased entry of CIA Test 1 and CIA Test 2 marks, assignment/quiz/lab marks, and ESE marks or an absent status.
- Generate a student semester mark sheet and reports for attendance defaulters and department-wise class performance.
- Use JDBC to connect the Java application to the MySQL database.

## FLOW CHART:



**PROCEDURE:**

- Database Creation: Create the kec_academic database in MySQL and define tables for students, courses, enrollments, attendance, and assessment marks.
- Sample Data Insertion: Insert sample student and course details into the database to test the system's functionalities.
- JDBC Configuration: Add the MySQL Connector/J library and configure the database connection details in DBConnection.java.
- Database Connection: Establish a connection between the Java application and MySQL using JDBC to store, retrieve, and update academic information.
- Student Management: Add new students with their roll numbers, names, departments, and semester details, and display existing student records.
- Course Management and Enrollment: Add course details, including course codes, titles, credits, and course types, and enroll students in the selected courses.
- Attendance Management: Record student attendance and maintain medical condonation details according to the application's configured rules.
- Assessment Mark Entry: Enter CIA 1, CIA 2, assignment, quiz, laboratory, and ESE marks, including examination absence details when required.
- Mark Sheet Generation: Retrieve student enrollment and assessment details to generate semester mark sheets and display academic performance.
- Attendance Defaulter Report: Identify students whose attendance falls below the configured threshold and generate a report to monitor attendance eligibility.
- Department Performance Summary: Analyze academic results and display department-wise performance and pass-percentage information supported by the application.
- Menu Navigation: Allow users to return to the main menu after each operation and select other functions until they choose to exit.

## DATABASE SCHEMA:

### 1.STUDENTS TABLE:

```
mysql> DESCRIBE students;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| roll_no          | varchar(20) | NO   | PRI | NULL    |       |
| student_name     | varchar(100)| NO   |     | NULL    |       |
| department       | varchar(20) | NO   |     | NULL    |       |
| current_semester | tinyint     | NO   |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

### 2. COURSES TABLE:

```
mysql> DESCRIBE courses;
+-------------+-----------------------------------------+------+-----+---------+-------+
| Field       | Type                                    | Null | Key | Default | Extra |
+-------------+-----------------------------------------+------+-----+---------+-------+
| course_code | varchar(20)                             | NO   | PRI | NULL    |       |
| title       | varchar(150)                            | NO   |     | NULL    |       |
| credits     | decimal(4,1)                            | NO   |     | NULL    |       |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED')   | NO   |     | NULL    |       |
+-------------+-----------------------------------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

### 3. ENROLLMENTS TABLE:

```
mysql> DESCRIBE enrollments;
+------------------------------+-------------+------+-----+---------+----------------+
| Field                        | Type        | Null | Key | Default | Extra          |
+------------------------------+-------------+------+-----+---------+----------------+
| enrollment_id                | bigint      | NO   | PRI | NULL    | auto_increment |
| roll_no                      | varchar(20) | NO   | MUL | NULL    |                |
| course_code                  | varchar(20) | NO   | MUL | NULL    |                |
| semester                     | tinyint     | NO   |     | NULL    |                |
| classes_held                 | int         | NO   |     | 0       |                |
| classes_attended             | int         | NO   |     | 0       |                |
| medical_condonation_approved | tinyint(1)  | NO   |     | 0       |                |
+------------------------------+-------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

### 4. ASSESSMENT MARKS TABLE:

```
mysql> DESCRIBE assessment_marks;
+----------------+-------------+------+-----+-------------------+-----------------------------------------------+
| Field          | Type        | Null | Key | Default           | Extra                                         |
+----------------+-------------+------+-----+-------------------+-----------------------------------------------+
| enrollment_id  | bigint      | NO   | PRI | NULL              |                                               |
| cia1           | decimal(5,2)| YES  |     | NULL              |                                               |
| cia2           | decimal(5,2)| YES  |     | NULL              |                                               |
| assignment_mark| decimal(5,2)| YES  |     | NULL              |                                               |
| quiz_mark      | decimal(5,2)| YES  |     | NULL              |                                               |
| lab_mark       | decimal(5,2)| YES  |     | NULL              |                                               |
| ese_mark       | decimal(5,2)| YES  |     | NULL              |                                               |
| ese_absent     | tinyint(1)  | NO   |     | 0                 |                                               |
| updated_at     | timestamp   | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+----------------+-------------+------+-----+-------------------+-----------------------------------------------+
9 rows in set (0.00 sec)
```

# SOURCE CODE:

## MAINAPP.JAVA:

```
import java.sql.SQLException;
import java.util.Scanner;
public class MainApp {
    private static final Scanner in = new Scanner(System.in);
    private static final StudentDAO studentDAO = new StudentDAO();
    private static final AcademicDAO academicDAO = new AcademicDAO();
 public static void main(String[] args) {
        System.out.println("============================================================");
        System.out.println(" KEC R2024 STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM");
        System.out.println("============================================================");
        try {
            DBConnection.getConnection().close();
            System.out.println("Database connection successful.");
        } catch (SQLException e) {
            System.out.println("Database connection failed: " + e.getMessage());
            System.out.println("Check MySQL service, schema, username/password and Connector/J.");
            return;
        }
while (true) {
            System.out.println("\nMAIN MENU");
            System.out.println("1. Add student");
            System.out.println("2. List students");
            System.out.println("3. Add course");
            System.out.println("4. List courses");
            System.out.println("5. Enroll student in course");
            System.out.println("6. Update attendance / medical condonation");
            System.out.println("7. Phase 1: Enter CIA Test 1 and CIA Test 2");
            System.out.println("8. Phase 2: Enter assignment / quiz / lab marks");
```

```
            System.out.println("9. Phase 3: Enter ESE marks or mark AB");
            System.out.println("10. Generate student semester mark sheet");
            System.out.println("11. Attendance defaulter report (<65%)");
            System.out.println("12. Department-wise class performance summary");
            System.out.println("0. Exit");
            int choice = readInt("Choose: ");
            try {
                switch (choice) {
                    case 1: addStudent(); break;
                    case 2: studentDAO.listStudents(); break;
                    case 3: addCourse(); break;
                    case 4: studentDAO.listCourses(); break;
                    case 5: enroll(); break;
                    case 6: attendance(); break;
                    case 7: phase1(); break;
                    case 8: phase2(); break;
                    case 9: phase3(); break;
                    case 10: academicDAO.printMarkSheet(readText("Roll number: ").toUpperCase(),
                                                        readInt("Semester: ")); break;
                    case 11: academicDAO.attendanceDefaulters(); break;
                    case 12: academicDAO.classSummary(); break;
                    case 0: System.out.println("Goodbye."); return;
                    default: System.out.println("Choose a listed option.");
                }
            } catch (SQLException | IllegalArgumentException e) {
                System.out.println("Operation failed: " + e.getMessage());
            }
        }
    }
    private static void addStudent() throws SQLException {
        String roll = readText("Roll number (e.g. 24ITR001): ").toUpperCase();
```

```
        String name = readText("Student name: ");
        String dept = readText("Department (CSE/ECE/IT/MECH/etc.): ").toUpperCase();
        int sem = readInt("Current semester (1-8): ");
        if (sem < 1 || sem > 8) throw new IllegalArgumentException("Semester must be 1-8.");
        studentDAO.addStudent(roll, name, dept, sem);
        System.out.println("Student added successfully.");
    }
    private static void addCourse() throws SQLException {
        String code = readText("Course code (e.g. 24ITR31): ").toUpperCase();
        String title = readText("Course title: ");
        double credits = readDouble("Credits (e.g. 3): ");
        String type = readText("Course type (THEORY/PRACTICAL/EMBEDDED): ").toUpperCase();
        if (!type.equals("THEORY") && !type.equals("PRACTICAL") && !type.equals("EMBEDDED"))
            throw new IllegalArgumentException("Invalid course type.");
        studentDAO.addCourse(code, title, credits, type);
        System.out.println("Course added successfully.");
    }
    private static void enroll() throws SQLException {
        String roll = readText("Roll number: ").toUpperCase();
        String code = readText("Course code: ").toUpperCase();
        int sem = readInt("Semester: ");
        academicDAO.enroll(roll, code, sem);
        System.out.println("Enrollment created.");
    }
    private static void attendance() throws SQLException {
        String roll = readText("Roll number: ").toUpperCase();
        String code = readText("Course code: ").toUpperCase();
        int sem = readInt("Semester: ");
        int held = readInt("Total classes held: ");
        int attended = readInt("Classes attended: ");
        boolean approved = readYesNo("Medical condonation approved? (y/n): ");
```

```
        academicDAO.updateAttendance(roll, code, sem, held, attended, approved);
        double pct = held == 0 ? 0 : 100.0 * attended / held;
        System.out.printf("Attendance saved: %.2f%%%n", pct);
        if (pct < 65) System.out.println("Rule: SA / barred from ESE.");
        else if (pct < 75 && !approved) System.out.println("Rule: ESE eligibility pending medical condonation approval.");
        else System.out.println("Rule: Eligible for ESE.");
    }
  private static void phase1() throws SQLException {
        String roll = readText("Roll number: ").toUpperCase();
        String code = readText("Course code: ").toUpperCase(); int sem = readInt("Semester: ");
        Double c1 = readMark("CIA Test 1 (0-100): ");
        Double c2 = readMark("CIA Test 2 (0-100): ");
        academicDAO.enterMarks(roll, code, sem, "PHASE1", c1, c2, null, null, null, null, false);
        System.out.println("Phase 1 marks saved.");
    }
private static void phase2() throws SQLException {
        String roll = readText("Roll number: ").toUpperCase();
        String code = readText("Course code: ").toUpperCase(); int sem = readInt("Semester: ");
        Double a = readOptionalMark("Assignment mark (0-100, blank to skip): ");
        Double q = readOptionalMark("Quiz mark (0-100, blank to skip): ");
        Double lab = readOptionalMark("Lab evaluation mark (0-100, blank to skip): ");
        academicDAO.enterMarks(roll, code, sem, "PHASE2", null, null, a, q, lab, null, false);
        System.out.println("Phase 2 marks saved.");
    }
private static void phase3() throws SQLException {
        String roll = readText("Roll number: ").toUpperCase();
        String code = readText("Course code: ").toUpperCase(); int sem = readInt("Semester: ");
        boolean absent = readYesNo("Was the student absent in ESE? (y/n): ");
        Double ese = absent ? null : readMark("ESE mark (0-100): ");
        academicDAO.enterMarks(roll, code, sem, "PHASE3", null, null, null, null, null, ese, absent);
```

```
        System.out.println(absent ? "ESE absence recorded as AB." : "ESE marks saved.");
    }
    private static String readText(String prompt) {
        System.out.print(prompt);
        String s = in.nextLine().trim();
        if (s.isEmpty()) throw new IllegalArgumentException("Input cannot be blank.");
        return s;
    }
    private static int readInt(String prompt) {
        try { return Integer.parseInt(readText(prompt)); }
        catch (NumberFormatException e) { throw new IllegalArgumentException("Enter a whole number."); }
    }
    private static double readDouble(String prompt) {
        try { return Double.parseDouble(readText(prompt)); }
        catch (NumberFormatException e) { throw new IllegalArgumentException("Enter a valid number."); }
    }
    private static Double readMark(String prompt) {
        double v = readDouble(prompt);
        if (v < 0 || v > 100) throw new IllegalArgumentException("Marks must be 0-100.");
        return v;
    }
    private static Double readOptionalMark(String prompt) {
        System.out.print(prompt);
        String s = in.nextLine().trim();
        if (s.isEmpty()) return null;
        try {
            double v = Double.parseDouble(s);
            if (v < 0 || v > 100) throw new IllegalArgumentException("Marks must be 0-100.");
            return v;
        } catch (NumberFormatException e) { throw new IllegalArgumentException("Enter a valid number or leave blank."); }
```

```
    }
    private static boolean readYesNo(String prompt) {
        String s = readText(prompt).toLowerCase();
        if (s.equals("y") || s.equals("yes")) return true;
        if (s.equals("n") || s.equals("no")) return false;
        throw new IllegalArgumentException("Enter y or n.");
    }
}
```

## DB CONNECTION.JAVA:

```
Import java.sql.Connection;
Import java.sql.DriverManager;
Import java.sql.SQLException;

Public final class DBConnection {
    Private static final String URL =
        System.getenv().getOrDefault("KEC_DB_URL",
            "jdbc:mysql://localhost:3306/kec_academic?useSSL=false&serverTimezone=UTC");
    Private static final String USER =
        System.getenv().getOrDefault("KEC_DB_USER", "root");
    Private static final String PASSWORD =
        System.getenv().getOrDefault("KEC_DB_PASSWORD", "Hrisehmesa-123");

    Private DBConnection() {}

    Public static Connection getConnection() throws SQLException {
        Return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

**OUTPUT SCREENSHOTS:**

**MAIN MENU:**

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose:
```

# STUDENT & COURSE MANAGEMENT:

## 1.STUDENT REGISTRATION:

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 1
Roll number (e.g. 24ITR001): 24ITR149
Student name: ravi R
Department (CSE/ECE/IT/MECH/etc.): ECE
Current semester (1-8): 2
Student added successfully.
```

## 2.STUDENTS LIST:

```
Choose: 2

ROLL NO       NAME                          DEPARTMENT  SEM
------------------------------------------------------------
24ITR001      Arun Kumar                    IT          3
24ITR002      Meena Devi                    IT          3
24ITR003      sahana                        IT          3
24ITR023      Kavitha M                     IT          3
24ITR024      Thenmozhi S                   IT          3
24ITR025      Kayalvizhi R                  IT          3
24ITR026      Nivetha K                     IT          3
24ITR027      Yazhini S                     IT          3
24ITR028      Mahalakshmi P                 IT          3
24ITR030      Tamilselvi M                  IT          3
```

## 3.COURSE ENROLLMENT:

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 3
Course code (e.g. 24ITR31): 24ITR51
Course title: Environmental Science
Credits (e.g. 3): 1
Course type (THEORY/PRACTICAL/EMBEDDED): THEORY
Course added successfully.
```

## 4.COURSE LIST:

```
Choose: 4

COURSE CODE   TITLE                                          CREDITS TYPE
-------------------------------------------------------------------------
24ITR31       Database Management Systems                    3.0     THEORY
24ITR32       Java Programming Laboratory                    2.0     PRACTICAL
24ITR33       Data Analytics Fundamentals                    3.0     EMBEDDED
24ITR34       Operating Systems                              3.0     THEORY
24ITR35       privacy and security                           2.0     THEORY
24ITR36       Computer Networks                              3.0     THEORY
24ITR37       Data Structures                                3.0     THEORY
24ITR38       Python Programming                             3.0     THEORY
24ITR39       Web Technology Laboratory                      2.0     PRACTICAL
24ITR40       Database Laboratory                            2.0     PRACTICAL
24ITR51       Environmental Science                          1.0     THEORY
```

## 5.ENROLL STUDENT IN COURSE

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 5
Roll number: 24ITR003
Course code: 24ITR51
Semester: 2
Enrollment created.
```

# 6.ATTENDANCE AND MEDICAL CONDONATION

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 6
Roll number: 24ITR003
Course code: 24ITR51
Semester: 2
Total classes held: 32
Classes attended: 30
Medical condonation approved? (y/n): n
Attendance saved: 93.75%
Rule: Eligible for ESE.
```

# INCREMENTAL MARKS ENTRY:

## 1.PHASE 1 (CIA MARKS):

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 7
Roll number: 24ITR003
Course code: 24ITR51
Semester: 2
CIA Test 1 (0-100): 94
CIA Test 2 (0-100): 92
Phase 1 marks saved.
```

## 2.PHASE 2 (ASSIGNMENT/QUIZ/LAB MARKS):

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 8
Roll number: 24ITR003
Course code: 24ITR51
Semester: 2
Assignment mark (0-100, blank to skip): 90
Quiz mark (0-100, blank to skip): 89
Lab evaluation mark (0-100, blank to skip): 95
Phase 2 marks saved.
```

## 3. PHASE 3 (END SEMESTER MARKS):

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 9
Roll number: 24ITR003
Course code: 24ITR51
Semester: 2
Was the student absent in ESE? (y/n): n
ESE mark (0-100): 94
ESE marks saved.
```

# ATTENDANCE & GRADING RULES:

## 1.STUDENT SEMESTER MARK SHEET:

```
Choose: 10
Roll number: 24ITR003
Semester: 2

================ KEC R2024 - SEMESTER MARK SHEET ================
Roll No: 24ITR003 | Semester: 2
-----------------------------------------------------------------
COURSE     TYPE       CIA    ESE    TOTAL   GRADE    STATUS
-----------------------------------------------------------------
24ITR31    THEORY      35.2   54.0    89.2 SA       BARRED: ATTENDANCE
24ITR32    PRACTICAL   54.8   36.8    91.6 SA       BARRED: ATTENDANCE
24ITR33    EMBEDDED    46.1   47.0    93.1 SA       BARRED: ATTENDANCE
24ITR34    THEORY      35.9   52.8    88.7 SA       BARRED: ATTENDANCE
24ITR35    THEORY       0.0    0.0     0.0 SA       BARRED: ATTENDANCE
24ITR51    THEORY      36.5   56.4    92.9 O        PASS
-----------------------------------------------------------------
SGPA: 0.71 | Equivalent percentage (SGPA x 10): 7.14%
Note: Pending assessments and non-passing grades contribute 0 grade points.
```

## 2.ATTENDANCE DEFAULTER REPORT:

```
  MAIN MENU
  1. Add student
  2. List students
  3. Add course
  4. List courses
  5. Enroll student in course
  6. Update attendance / medical condonation
  7. Phase 1: Enter CIA Test 1 and CIA Test 2
  8. Phase 2: Enter assignment / quiz / lab marks
  9. Phase 3: Enter ESE marks or mark AB
  10. Generate student semester mark sheet
  11. Attendance defaulter report (<65%)
  12. Department-wise class performance summary
  0. Exit
  Choose: 11

  ATTENDANCE DEFAULTERS (BELOW 65%)
  ROLL NO       NAME                    COURSE    ATTENDANCE GRADE
  24ITR002      Meena Devi              24ITR31      60.00% SA
```

## DEPARTMENT WISE CLASS PERFORMANCE SUMMARY:

```
MAIN MENU
1. Add student
2. List students
3. Add course
4. List courses
5. Enroll student in course
6. Update attendance / medical condonation
7. Phase 1: Enter CIA Test 1 and CIA Test 2
8. Phase 2: Enter assignment / quiz / lab marks
9. Phase 3: Enter ESE marks or mark AB
10. Generate student semester mark sheet
11. Attendance defaulter report (<65%)
12. Department-wise class performance summary
0. Exit
Choose: 12
\nDEPARTMENT-WISE RESULT SUMMARY
DEPARTMENT    ENROLLMENTS  PASS        FAIL        AVG SGPA
IT            12           4           0           2.24
```

# DATABASE:

## STUDENTS DATABASE:

```
mysql> SELECT * FROM students;
+----------+--------------+------------+------------------+
| roll_no  | student_name | department | current_semester |
+----------+--------------+------------+------------------+
| 24ITR001 | Arun Kumar   | IT         |                3 |
| 24ITR002 | Meena Devi   | IT         |                3 |
| 24ITR003 | sahana       | IT         |                3 |
| 24ITR023 | Kavitha M    | IT         |                3 |
| 24ITR024 | Thenmozhi S  | IT         |                3 |
| 24ITR025 | Kayalvizhi R | IT         |                3 |
| 24ITR026 | Nivetha K    | IT         |                3 |
| 24ITR027 | Yazhini S    | IT         |                3 |
| 24ITR028 | Mahalakshmi P | IT         |                3 |
| 24ITR030 | Tamilselvi M | IT         |                3 |
| 24ITR031 | Ilakkiya S   | IT         |                3 |
| 24ITR032 | Kanishka R   | IT         |                3 |
| 24ITR033 | Pavithra K   | IT         |                3 |
| 24ITR036 | Sowmiya R    | IT         |                3 |
| 24ITR046 | Dinesh Kumar | IT         |                3 |
| 24ITR047 | Sathish Raj  | IT         |                3 |
| 24ITR048 | Madhan Kumar | IT         |                3 |
| 24ITR050 | Ashwin Raj   | IT         |                3 |
| 24ITR053 | Kavin Raj    | IT         |                3 |
| 24ITR055 | Manikandan S | IT         |                3 |
| 24ITR121 | shalini      | ECE        |                4 |
| 24ITR149 | ravi R       | ECE        |                2 |
| 25ITR125 | sangavi      | CSE        |                2 |
+----------+--------------+------------+------------------+
23 rows in set (0.00 sec)
```

## COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+----------------------------+---------+-------------+
| course_code | title                      | credits | course_type |
+-------------+----------------------------+---------+-------------+
| 24ITR31     | Database Management Systems |     3.0 | THEORY      |
| 24ITR32     | Java Programming Laboratory |     2.0 | PRACTICAL   |
| 24ITR33     | Data Analytics Fundamentals |     3.0 | EMBEDDED    |
| 24ITR34     | Operating Systems          |     3.0 | THEORY      |
| 24ITR35     | privacy and security       |     2.0 | THEORY      |
| 24ITR36     | Computer Networks          |     3.0 | THEORY      |
| 24ITR37     | Data Structures            |     3.0 | THEORY      |
| 24ITR38     | Python Programming         |     3.0 | THEORY      |
| 24ITR39     | Web Technology Laboratory  |     2.0 | PRACTICAL   |
| 24ITR40     | Database Laboratory        |     2.0 | PRACTICAL   |
| 24ITR51     | Environmental Science      |     1.0 | THEORY      |
+-------------+----------------------------+---------+-------------+
11 rows in set (0.00 sec)
```

**ENROLLMENTS DATABASE:**

```
mysql> SELECT * FROM enrollments;
+--------------+----------+-------------+----------+-------------+-----------------+----------------------------+
| enrollment_id | roll_no  | course_code | semester | classes_held | classes_attended | medical_condonation_approved |
+--------------+----------+-------------+----------+-------------+-----------------+----------------------------+
|            1 | 24ITR001 | 24ITR31     |        3 |          40 |              34 |                          0 |
|            2 | 24ITR001 | 24ITR32     |        3 |          40 |              36 |                          0 |
|            3 | 24ITR001 | 24ITR33     |        3 |          40 |              32 |                          0 |
|            4 | 24ITR002 | 24ITR31     |        3 |          40 |              24 |                          0 |
|            8 | 24ITR002 | 24ITR32     |        3 |           0 |               0 |                          0 |
|            9 | 24ITR003 | 24ITR32     |        3 |          32 |              29 |                          0 |
|           10 | 24ITR003 | 24ITR51     |        2 |          32 |              30 |                          0 |
|           11 | 24ITR003 | 24ITR31     |        2 |           0 |               0 |                          0 |
|           12 | 24ITR003 | 24ITR32     |        2 |           0 |               0 |                          0 |
|           13 | 24ITR003 | 24ITR33     |        2 |           0 |               0 |                          0 |
|           14 | 24ITR003 | 24ITR34     |        2 |           0 |               0 |                          0 |
|           15 | 24ITR003 | 24ITR35     |        2 |           0 |               0 |                          0 |
+--------------+----------+-------------+----------+-------------+-----------------+----------------------------+
12 rows in set (0.00 sec)
```

**ASSESSMENT MARKS DATABASE:**

```
mysql> SELECT * FROM assessment_marks;
+--------------+-------+-------+----------------+-----------+----------+----------+------------+---------------------+
| enrollment_id | cia1  | cia2  | assignment_mark | quiz_mark | lab_mark | ese_mark | ese_absent | updated_at          |
+--------------+-------+-------+----------------+-----------+----------+----------+------------+---------------------+
|            1 | 82.00 | 78.00 |          85.00 |     80.00 |     NULL |    76.00 |          0 | 2026-10-24 22:04:53 |
|            2 | 90.00 | 88.00 |           NULL |      NULL |    92.00 |    86.00 |          0 | 2026-10-24 22:04:53 |
|            3 | 75.00 | 80.00 |          82.00 |     79.00 |    84.00 |    73.00 |          0 | 2026-10-24 22:04:53 |
|            4 | 60.00 | 55.00 |          62.00 |     58.00 |     NULL |    40.00 |          0 | 2026-10-24 22:04:53 |
|            8 |  NULL |  NULL |           NULL |      NULL |     NULL |     NULL |          0 | 2026-10-24 22:25:29 |
|            9 | 90.00 | 88.00 |          92.00 |     95.00 |    94.00 |     NULL |          0 | 2026-10-24 22:42:14 |
|           10 | 94.00 | 92.00 |          90.00 |     89.00 |    95.00 |    94.00 |          0 | 2026-10-26 10:43:16 |
|           11 |  NULL |  NULL |          84.00 |     92.00 |    90.00 |    90.00 |          0 | 2026-10-26 10:57:33 |
|           12 | 94.00 | 90.00 |          94.00 |     85.00 |    90.00 |    92.00 |          0 | 2026-10-26 10:57:49 |
|           13 | 87.00 | 94.00 |          90.00 |     94.00 |    96.00 |    94.00 |          0 | 2026-10-26 10:58:15 |
|           14 | 89.00 | 92.00 |          90.00 |     88.00 |    94.00 |    88.00 |          0 | 2026-10-26 10:58:36 |
|           15 |  NULL |  NULL |           NULL |      NULL |     NULL |     NULL |          0 | 2026-10-26 10:48:47 |
+--------------+-------+-------+----------------+-----------+----------+----------+------------+---------------------+
12 rows in set (0.00 sec)
```

**RESULT:**

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous), Regulation 2024, was developed using Java, JDBC, and MySQL. The system facilitates the management of student records, course details, and assessment marks. It aims to simplify academic evaluation, reduce manual effort, and improve the accuracy and efficiency of academic record management.

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**GITHUB:**

| | |
|---|---|
| **LINK:** | **https://github.com/Sahana-2373/KEC_R2024_Academic_System** |